Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

1

Questions and Exercises

① State the kinetic theory of ideal gases.

② Write the equation of an ideal gas.

③ What is a thermodynamic process? A cyclic process? A non-cyclic process?

⑤ A $2.00\ ml$ volume container contains $50\ \text{mg}$ of gas at a pressure of $100\ \text{kPa}$. The mass of each gas particle is $8.0\times10^{-26}\ \text{kg}$.

(a) What is the number of moles of the gas?

(b) Find the average translational kinetic energy of each particle.

⑥ Calculate the work done by the gas in the following cases:

(a) A gas expands at a constant pressure of $1.50 \times 10^3\ \text{Pa}$, increasing its volume from $75 \times 10^4\ \text{cm}^3$ to $90 \times 10^4\ \text{cm}^3$.

(b) A gas expands at a constant pressure of $200\ \text{kPa}$, increasing its volume from $0.75\ \text{m}^3$ to $1.9\ \text{m}^3$.

(c) A gas expands volume from $0.40\ \text{m}^3$ to $0.60\ \text{m}^3$ while the pressure increases from $1.00 \times 10^3\ \text{Pa}$ to $220 \times 10^3\ \text{Pa}$.

(d) A gas is compressed as the pressure increases from $1 \times 10^5\ \text{Pa}$ to $2.50 \times 10^5\ \text{Pa}$, while its volume decreases from $0.50\ \text{m}^3$ to $0.70\ \text{m}^3$.

⑦ A cylinder contains $0.5\ \text{moles}$ of an ideal monoatomic gas at a temperature of $310\ \text{K}$. The volume expands from $310\ \text{dm}^3$ to $450\ \text{dm}^3$, where the temperature is kept constant.

(a) Calculate the work done by the system during this volume change.

(b) Calculate the change in internal energy of the gas.

(c) Calculate the amount of energy absorbed by the system during the volume change.

⑧ A Carnot engine has a power output of $150\ \text{kW}$. The engine operates between two reservoirs at $20.0°\text{C}$ and $500°\text{C}$.

(a) How much energy does it take in per hour?

(b) How much energy is lost per hour in its exhaust?

⑨ An ion of ^{58}Ni (with charge $+e$ and mass $9.62 \times 10^{-26}\ \text{kg}$) is accelerated under a potential difference of $3000\ \text{V}$. The ion is deflected in a magnetic field of $0.12\ \text{T}$.

(a) Calculate the radius of the ^{58}Ni ion.

(b) Calculate the difference in the radius between ^{58}Ni and ^{60}Ni, assuming that the mass ratio is $58{:}60$.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination:** .....

Room N°: ..... **Seat N°:** ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

2

Questions and Exercises

① What is a mole, Avogadro's number, and atomic unit?

② What does the first law of thermodynamics study? State this law.

③ A cylinder contains a mixture of helium and argon gas in equilibrium at 150 °C. Given: Molar mass of helium and argon is $M_{He} = 4\ \text{g/mole}$, and $M_{Ar} = 40\ \text{g/mole}$, respectively.

(a) What is the average kinetic energy for each type of gas molecule?

(b) What is the rms speed of each type of molecule?

④ (a) A 1.0-mole sample of an ideal gas is kept at 0.0 °C during an expansion from 3.0 L to 10.0 L.

How much work is done by the gas during the expansion? Given: $\ln 3 = 1.09$, $\ln 10 = 2.30$

(b) An ideal gas at a temperature of $T = 300\ \text{K}$ during an expansion from $V_1 = 0.31\ \text{m}^3$ to $V_2 = 0.45\ \text{m}^3$. The gas contains $n = 0.5$ moles.

How much work done by the gas during the expansion? Given: $\ln 0.31 = -1.17$, $\ln 0.45 = 0.79$

⑤ Pisey is jogging along the beach, performing 4.3×10^5 J of work and released 3.8×10^5 J of heat energy.

(a) What is the change in Pisey's internal energy?

(b) If she changes from jogging to walking, she released 1.2×10^5 J of heat and the internal energy decreased by 2.6×10^5 J. How much work is done by Pisey during walking?

⑥ Imagine a Carnot engine that operates between the temperatures $T_h = 850\ \text{K}$ and $T_c = 300\ \text{K}$. The engine performs 1200 J of work each cycle, which takes 0.25 s.

(a) What is the efficiency of this engine?

(b) What is the average power of this engine?

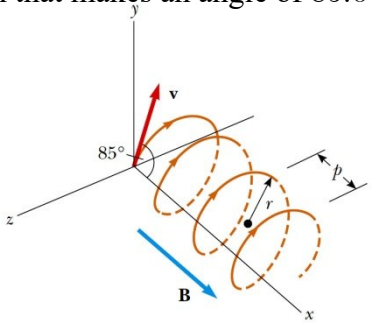
(c) How much energy Q_h is extracted as heat from the high-temperature reservoir every cycle?

(d) How much energy Q_c is delivered as heat to the low-temperature reservoir every cycle?

⑦ A uniform magnetic field of magnitude 0.150 T is directed along the positive x axis. A positron moving at $5.00 \times 10^6\ \text{m/s}$ enters the field along a direction that makes an angle of 85.0° with the x axis. The motion of the particle is expected to be a helix. Calculate

(a) the pitch p and

(b) the radius r of the trajectory.



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

3

Questions and Exercises

① What is a system, state of a system, and thermodynamic system?

② State the principle initial and final states of a system. What type of process governs this transition?

③ Write the formula of total translational kinetic energy of the gas molecules.

④ Hydrogen gas is filled in a balloon at a volume of $100\ \text{cm}^3$, a temperature of $7°\text{C}$, and a pressure of $0.7\ \text{atm}$. Determine the number of hydrogen tubes required to fully inflate the balloon, assuming each hydrogen tube has a volume of $20\ \text{L}$ at a pressure of $150\ \text{atm}$ and a temperature of $27°\text{C}$. Assume the gas in the tube is completely transferred to the balloon.

⑤ A gas is contained in a tank with a pressure of $1.6 \times 10^6\ \text{Pa}$ and a volume of $4.0\ \text{m}^3$.

(a) What is the work done by the gas when it is compressed at constant pressure to twice its initial volume?

(b) What is the work done on the gas when it is compressed at constant pressure to one-fourth of its initial volume?

⑥ A cylinder with a cross-sectional area $A = 500\ \text{cm}^2$ contains ideal gas and is sealed by a movable piston. Heat is slowly supported to the gas while maintaining a constant pressure of $P = 10^5\ \text{Pa}$, causing the piston move vertically by a distance $10\ \text{cm}$.

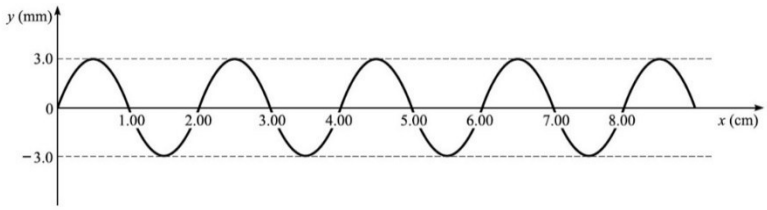
(a) What is this process called?

(b) How much work is done by the gas during this process?

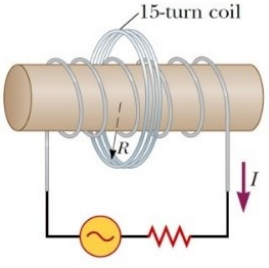
(c) If $600\ \text{J}$ of heat is supplied to the gas during this process, what is the change in internal energy of the gas?

⑦ Steam flows through the cylinder of a steam engine at constant pressure, performing $0.84\ \text{J}$ of work on the piston. The piston has a diameter of $1.6\ \text{m}$, and it moves a distance of $2.1\ \text{cm}$. What is the pressure exerted by the steam?

⑧ The wave shown in Fig is being sent out by a $60\ \text{cycle/s}$ vibrator. Find the following for the wave: (a) amplitude, (b) frequency, (c) wavelength, (d) speed, (e) period.



⑨ A coil of $15\ \text{turns}$ and radius $10.0\ \text{cm}$ surrounds a long solenoid of radius $2.00\ \text{cm}$ and $1.00 \times 10^3\ \text{turns/meter}$ (Fig). The current in the solenoid changes as $I = 5.00 \sin 120t$, where I is in amperes and t is in seconds. Find the induced emf in the 15-turn coil as a function of time.



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination:** .....

Room N°: ..... **Seat N°:** ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

4

Questions and Exercises

① Describe the process of the Carnot cycle.

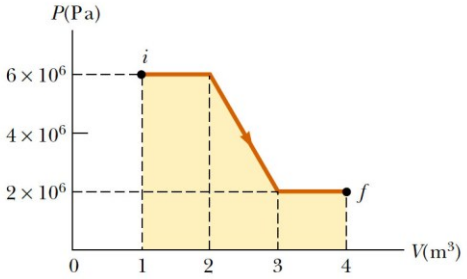
② What is the difference between an internal combustion engine and an external combustion engine?

③ What is a theoretical (ideal) engine?

④ A container in the shape of a cube $10.0\ \text{cm}$ on each edge contains air (with equivalent molar mass $28.9\ \text{g/mol}$) at atmospheric pressure and temperature $300\ \text{K}$. Find

(a) the mass of the gas,
(b) the gravitational force exerted on it, and
(c) the force it exerts on each face of the cube. Take gravity $g = 9.80\ \text{m/s}^2$.
(d) Why does such a small sample exert such a great force?

⑤ (a) Determine the work done on a fluid that expands from i to f as indicated in Figure.



(b) What If? How much work is performed on the fluid if it is compressed from f to i along the same path?

⑥ In each of the following situations, find the change in internal energy of the system.

(a) A system absorbs $500\ \text{cal}$ of heat and at the same time does $400\ \text{J}$ of work.

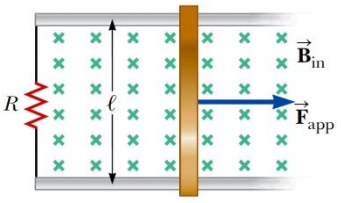
(b) A system absorbs $300\ \text{cal}$ and at the same time $420\ \text{J}$ of work is done on it.

© Twelve hundred calories is removed from a gas at constant volume. Give you answers in kilojoules.

⑦ A car engine receives $150\ \text{J}$ of heat from gasoline-air mixture every second and expels $120\ \text{kJ}$ of heat to the atmosphere.

(a) What is the mechanical work done during $5\ \text{minutes}$?
(b) What is the thermal efficiency of the engine?
(c) If the mechanical efficiency is 92%, how much work is done and useful efficiency by the crankshaft?

⑧ Figure shows a top view of a bar that can slide on two frictionless rails.



The resistor is $\text{R} = 6.00\ \Omega$, and a 2.50-T magnetic field is directed perpendicularly downward, into the paper. Let, $l = 1.20\ \text{m}$.

(a) Calculate the applied force required to move the bar to the right at a constant speed of $2.00\ \text{m/s}$.
(b) At what rate is energy delivered to the resistor?

⑨ A solenoid with $2000\ \text{turns}$ per unit length and a cross-sectional area of $\pi\ \text{m}^2$ carries a time-varying current given by $i\ (\text{t}) = 20 - 2\text{t}^2$ (A Determine the induced electromotive force (emf) at the instant $t = 0.2\text{s}$.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

5

Questions and Exercises

① Write the formula for the work done by the gas during volume expansion, sketch a PV diagram to illustrate the work done by the gas, and describe the thermodynamic process of the gas in the following cases

(a) Constant pressure (b) Constant temperature (c) Constant volume (d) changing pressure and volume.

② What is the isobaric process, isothermal process, isochoric process and adiabatic process?

③ Given that the root mean square (rms) speed of a helium atom at a certain temperature is $1350\ \mathrm{m/s}$, determine the rms speed of an oxygen molecule at this temperature. The molar mass of O_2 is 32g/mol and for He is $4\mathrm{g/mol}$.

④ In a cylinder contains helium gas with a mass of $6.4\ \mathrm{g}$ and a temperature of $27\ ^\circ\mathrm{C}$. The gas undergoes an isothermal expansion, with the temperature held constant, as the volume increases from $4\ \mathrm{L}$ to $12\ \mathrm{L}$.

Given: $\ln 3 = 1.098$, $\ln 2 = 2.484$.

(a) Calculate the work done by the gas during the volume expansion.

(b) Determine the change in the internal energy of the gas.

(c) Calculate the amount of heat absorbed by the system during this process.

⑤ An ideal Carnot engine takes heat from a source at $317\ ^\circ\mathrm{C}$, does some external work and delivers the remaining energy to a heat sink at $117\ ^\circ\mathrm{C}$. How much work is done by the engine and heat delivered to the heat sink, if $500\ \mathrm{kcal}$ of heat is taken from the source, then the heat delivered to the heat sink?

⑥ An object undergoes oscillations due to two waves propagating in the same direction with identical frequencies. The equations describing the oscillations are:

$y_1 = 2\sin(\pi t + \pi/4)\ (\mathrm{cm})$, and $y_2 = 2\sin(\pi t + \pi/3)\ (\mathrm{cm})$

(a) Determine the period, frequency, and phase difference between the two oscillations.

(b) Calculate the amplitude and phase angle of the resultant wave.

⑦ An alpha particle $\alpha(\mathrm{He}^{2+})$ (mass $m = 6.7 \times 10^{-27}\ \mathrm{kg}$) is accelerated from rest across a potential difference of $1900\ \mathrm{V}$. It then moves into a uniform magnetic field of magnitude $B = 0.34\ \mathrm{T}$, which is perpendicular to its velocity.

(a) Calculate the radius of the particle's path in the magnetic field.

(b) Calculate the period of the circular path.

⑧ In a model of an AC generator, a 510-turn rectangular coil of wire with dimensions $0.082\ \mathrm{m}$ and $0.25\ \mathrm{m}$ rotates at an angular frequency of $12.8\ \mathrm{rad/s}$ in a uniform magnetic field of magnitude $0.65\ \mathrm{T}$. Find the maximum induced emf in the coil.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

6

Questions and Exercises

① Write the wave equation as a function of time and distance.

② What is a wavelength?

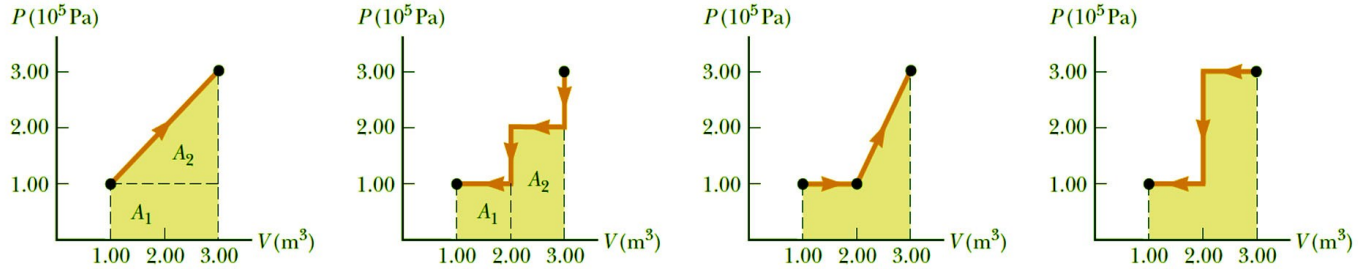
③ What is a standing wave? Give an example related to the standing wave equation.

④ (a) How many atoms of helium gas fill a balloon having a diameter of $30.0\ \text{cm}$ at $20.0\ ^\circ\text{C}$ and $1.00\ \text{atm}$?

(b) What is the average kinetic energy of the helium atoms?

(c) What is the root-mean square of the helium atom?

⑤ Given that PV diagram as shown in Figure below. Calculate the work done by the gas in the following cases:



⑥ A heat engine does $250\ \text{J}$ of work while the internal energy decreases by $500\ \text{J}$. What is the heat energy transferred during this process?

⑦ Calculate the change in internal energy for each of the following scenarios:

(a) A system absorbs $2000\ \text{J}$ of heat and has $400\ \text{J}$ of work done on it.

(b) A system absorbs $1200\ \text{J}$ of heat and has $400\ \text{J}$ of work done on it.

(c) $300\ \text{J}$ of heat is released from the system during a process at constant volume.

⑧ A Carnot engine absorbs $1200\ \text{cal}$ of heat in one cycle and operates between a hot reservoir at $500\ \text{K}$ and a cold reservoir at $300\ \text{K}$.

(a) Calculate the efficiency of the heat engine.

(b) Determine the heat expelled by the engine.

(c) Calculate the work done by the engine in one cycle in Joules.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

7

Questions and Exercises

① What is the superposition of a wave?

② What is a transverse wave and a longitudinal wave?

③ What is the phenomenon of resonance?

④ The pressure of a gas in a 100 mL container is 200 kPa and the average translation kinetic energy of each gas particle is 6×10^{-21} J.

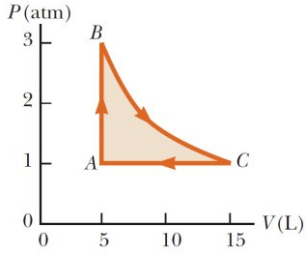
(a) Find the number of gas particles in the container.

(b) How many moles are there in the container?

(c) Determine the density of the gas particles.

⑤ At what temperature is the rms velocity of a hydrogen molecule equal to that of an oxygen molecule equal to that of an oxygen molecule at 47 °C? Given: $M_H = 2$ g/mol, $M_{O2} = 32$ g/mol.

⑥ A sample of a monatomic ideal gas occupies 5.00 L at atmospheric pressure and 300 K (point A in Fig). It is warmed at constant volume to 3.00 atm (point B). Then it is allowed to expand isothermally to 1.00 atm (point C) and at last compressed isobarically to its original state.



(a) Find the number of moles in the sample.

(b) Find the temperature at point A, B and C.

(c) Find the volume at point C.

(d) Find the network in a cyclic process ABCA.

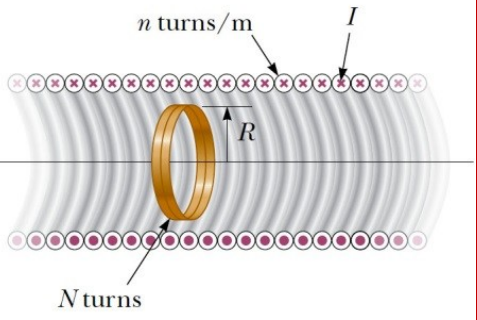
⑦ Heat engine X takes in four times more energy by heat from the hot reservoir than heat engine Y. Engine X delivers two times more work, and it rejects seven times more energy by heat to the cold reservoir than heat engine Y. Find the efficiency of (a) heat engine X and (b) heat engine Y.

⑧ An object oscillates due to two oscillators that have the same directions of propagation and frequencies: $y_1 = 2\sin(3\pi x - \pi/6)$, and $y_2 = 10\sin(3\pi x + \pi/2)$ where x and y are measured in centimeters, and t is measured in seconds.

(a) Determine the period, frequency and phase difference of two waves.

(b) Determine the amplitude and constant phase of a resultant wave.

⑨ A long solenoid has $n = 400$ turns per meter and carries a current given by $I = 30.0(1 - e^{-1.60t})$, where I is in amperes and t is in seconds. Inside the solenoid and coaxial with it is a coil that has a radius of R = 6.00 cm and consists of a total of N = 250 turns of fine wire (Fig). What emf is induced in the coil by the changing current?



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

8

Questions and Exercises

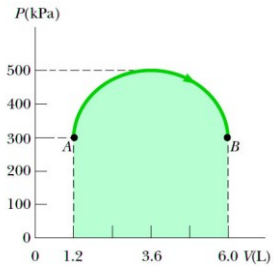
① What is the magnetic field?

② What is a uniform magnetic field? Where does this field occur?

③ What is the inductance of a circuit? What factors does it depend on, and what is its SI unit?

④ A glass bulb of volume $400\ \text{cm}^3$ is connected to another bulb of volume $200\ \text{cm}^3$ by means of a tube of negligible volume. The bulbs contain dry air and are both at a common temperature and pressure of $20\ °\text{C}$ and $1.000\ \text{atm}$, respectively. The larger bulb is immersed in steam at $100\ °\text{C}$ and the smaller in melting ice at $0\ °\text{C}$. Find the final common pressure.

⑤ A sample of an ideal gas is in a vertical cylinder fitted with a piston. As $5.79\ \text{kJ}$ of energy is transferred to the gas by heat to raise its temperature, the weight on the piston is adjusted so that the state of the gas changes from point A to point B along the semicircle shown in Figure. Find the change in internal energy of the gas.



⑥ An actual engine operates between hot sinks at $500\ \text{K}$ and $300\ \text{K}$. It releases $500\ \text{kJ}$ of heat from hot sink and does $150\ \text{kJ}$ of work per cycle.

(a) What is the thermal efficiency of an ideal the engine?

(b) What is the thermal efficiency of the actual engine?

⑦ A flat coil has $N = 100\ \text{turns}$ and carries a current $I = 10\ \text{A}$. The loop has an average radius of $R = 20\ \text{cm}$. Calculate the magnitude of the magnetic induction at the center of the coil if the core is made of a metal with relative permeability $\mu_r = 1000$.

⑧ A $1.8\ \text{m}$ long is wound into a coil with a radius of $3.0\ \text{cm}$. If this coil is rotated at $100\ \text{rev/min}$ in a $0.05\ \text{T}$ magnetic field. What is its maximum emf?

⑨ An ac-circuit has an angular frequency ω and a frequency $f = 50\ \text{Hz}$. Assume rms voltage $V_{AB} = 120\ \text{V}$, $V_L = 160\ \text{V}$, $V_C = 56\ \text{V}$ and a rms current of $I = 0.2\ \text{A}$.

(a) What is impedances of the inductor Z_L, the capacitor Z_C and the circuit Z_{AB}?

(b) What is the resistance of the inductor, R?

(c) What is the inductance of the coil, L?

(d) What is the capacitance of the capacitor, C?

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

9

Questions and Exercises

① What is the phenomenon of electromagnetic induction?

② What is the magnetic flux?

③ What is the induce current?

④ A vessel of volume 20 L contains a mixture of hydrogen at a temperature of $T = 27\ ^\circ\text{C}$ and a pressure $P = 20\ \text{atm}$, molar mass of hydrogen $M_{O2} = 2\ \text{g/mol}$. Calculate the mass of hydrogen gas in the balloon.

⑤ Calculate the root-mean-square speed (v_{rms}) of a nitrogen molecule (N_2) with a molar mass of $M = 28\ \text{g/mole}$ at a temperature of 20°C. Also, determine the temperature at which v_{rms} becomes:

(a) Half of its initial value.

(b) Twice its initial value.

⑥ Gas in a container is at a pressure of $1.5\ \text{atm}$ and a volume of $4.0\ \text{m}^3$. What is the work done on the gas

(a) if it expands at constant pressure to twice its initial volume, and

(b) if it is compressed at constant pressure to one-quarter its initial volume?

⑦ A Carnot engine operates with a cold reservoir at 7°C and an efficiency of 50%. To what temperature must the hot reservoir be increased to raise the efficiency to 70%?

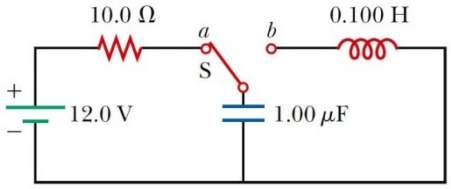
⑧ The switch in Figure is connected to position a for a long-time interval. At $t = 0$, the switch is thrown to position b. After this time, what are

(a) the frequency of oscillation of the LC circuit,

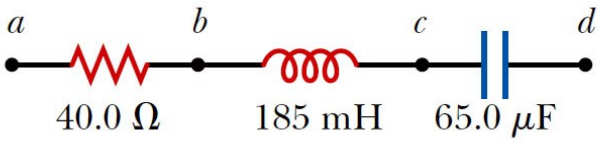
(b) the maximum charge that appears on the capacitor,

(c) the maximum current in the inductor, and

(d) the total energy the circuit possesses at $t = 3.00\ \text{s}$?



⑨ An AC source with $\Delta V_{\text{max}} = 150\ \text{V}$ and $f = 50.0\ \text{Hz}$ is connected between points a and d in Figure. Calculate the maximum voltages between (a) points a and b, (b) points b and c, (c) points c and d, and (d) points b and d.



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

10

Questions and Exercises

① What is one horsepower $(1\ \text{hp})$?

② What is the inductance? How can inductance become zero?

③ What is alternating current?

④ State the Lenz's law to determine the direction of induce current.

⑤ What is an electric motor?

⑥ A flat coil has N turns and an average radius $R = 5\ \text{cm}$. A current of $500\ \text{mA}$ flows through the coil, and the magnetic field of the coil is measured to be $B = 12.56 \times 10^{-4}\ \text{T}$. Calculate the number of turns of the coil.

⑦ A vibrator produces a wave oscillation with a frequency of $f = 20\ \text{Hz}$ and an amplitude $\text{A} = 5\ \text{cm}$. At the initial instant, the displacement has its maximum value.

(a) What is the angular frequency of the vibration?

(b) Write the wave equation as a function of time.

⑧ During each cycle, the engine expels $2430\ \text{J}$ of energy to the surroundings. The efficiency of the engine in one cycle is 46%.

(a) Calculate the amount of heat needed by the engine during each cycle.

(b) Calculate the power of the engine during one cycle if the cycle takes $2\ \text{ms}$.

⑨ A solenoid with cross-sectional area of $20\ \text{cm}^2$ is $2.512\ \text{m}$ and has $10^4\ \text{turns}$ and resistance $2\ \Omega$.

(a) Calculate the self-induction of the solenoid.

(b) The varying current through the solenoid is $i = 3\text{t}^2\ (\text{A})$. Write the expression for the voltage across the solenoid.

(c) If direct current of $12\ \text{V}$ applied to the terminal of the solenoid, calculate the current at instant $t = \tau$. Given: $\mu_0 = 4\pi \times 10^{-7}\ \text{H/m}$.

⑩ $1204.95\ \text{J}$ of work is done on $0.5\ \text{moles}$ ideal monoatomic gas at a temperature $290\ \text{K}$. The initial volume of the gas in the box with dimension $2\ \text{cm} \times 2\ \text{cm} \times 5\ \text{cm}$.

(a) Calculate the final volume of the gas.

(b) Calculate the internal energy of the gas.

(c) How much heat is absorbs or releases? Given: $\ln\ (0.37) = -1$

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

11

Questions and Exercises

① State the right-hand rule for a current in a circular coil.

② What is the phenomenon of the electromagnetic induction?

③ What is electrical oscillation? How many types of electrical oscillation exit, what are they?

④ Calculate the temperature in degrees Celsius of an ideal monatomic gas when the average kinetic energy per molecule is $K_{av} = 7.452 \times 10^{-21}$ J. The Boltzmann constant is given as $k_B = 1.38 \times 10^{-23}$ J/K.

⑤ Two long, straight, parallel wires carry equal currents, $I_1 = I_2$, and have a length of 2 cm. When currents I_1 and I_2 flow through the wires, they experience a magnetic interaction force of 16×10^{-5} N.

(a) If the currents flow in the same direction, describe the nature of the interaction force.

(b) Determine the magnitude of the current in each wire if the separation between them is $a = 4$ cm.

Given: $\mu_0 = 4\pi \times 10^{-7}$ Tm/A.

⑥ The equation of standing wave in a stretched string is given by: $y = (10\text{ cm})\sin(\pi/6x)\cos(30\pi t)$ (cm).

(a) What is the wave speed?

(b) Find the position of notes and antinodes.

(c) Determine the amplitude at $x = 2$ cm.

⑦ A coil of length l has a 100 turns per meter and an average radius of 10 cm. The coil has a self-inductance of 32.0 mH.

(a) Determine the total number of turns in the coil.

(b) Calculate the length of the coil. Given: $\mu_0 = 4\pi \times 10^{-7}$ Tm/A and $\pi^2 = 10$.

⑧ A 1-mole monoatomic ideal gas undergoes heating at constant pressure, with its temperature increasing from 300 K to 600 K.

(a) Calculate the work done by the gas during this process.

(b) Calculate the change in internal energy of the gas.

(c) Calculate the amount of heat energy added to the gas to achieve the temperature increase.

⑨ A coil has 100 turns, and the magnetic flux through it changes at a rate of 2 m Wb/s.

(a) Calculate the induced (emf) at the ends of the coil.

(b) Determine the induced current in the coil if its resistance is 0.40 Ω.

(c) Calculate the power dissipated due to Joule heating.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

12

Questions and Exercises

① What is the rms current of an alternating current (AC) circuit?

② State the right-hand rule for determining the direction of the magnetic force on a current-carrying wire placed in a uniform magnetic field.

③ An enclosed container with a volume of $20\ \text{mL}$ contains a gas at a pressure equal to atmospheric pressure $1\ \text{atm}$. The gas consists of $2 \times 10^{21}\ \text{molecules}$ moving randomly with velocity v.

④ In the same medium, two oscillators with the same frequency are described by the equations: $y_1 = (2\text{dm})\cos(10\pi t + \pi/2)$ and $y_2 = (5\text{dm})\sin(10\pi t)$, Determine the resulting equation describing the combination of these oscillations.

⑥ A Carnot engine performs work between a hot reservoir at temperature T_H and a cold reservoir at temperature T_C. The engine has a thermal efficiency of 66%.

(a) Find the ratio of the temperatures of the two reservoirs.

(b) Assuming the engine does $290\ \text{J}$ of work per cycle, calculate the amount of heat absorbed by the engine from the hot reservoir.

(c) Calculate the amount of heat released to the cold reservoir.

⑦ A current-carrying wire with a current of $10\ \text{A}$ is placed between the poles of a U-shaped magnet, where the magnetic induction is $50\ \text{mT}$. Assuming the wire is perpendicular to the magnetic field, it experiences a magnetic force with a magnitude of $0.25\ \text{N}$.

(a) Calculate the length of the wire (in centimeters).

(b) If the wire has a length of $2\ \text{cm}$, calculate the work done in displacing the current-carrying wire.

(c) Determine the magnetic flux produced by the area swept by the current-carrying wire.

⑧ A transformer has a primary coil connected to an AC voltage with a frequency of $50\ \text{Hz}$, an rms voltage of $110\ \text{V}$, and an rms current of $10\ \text{A}$. In the secondary coil, the rms current is measured to be $4.5\ \text{A}$ and is connected in series with a resistor of resistance $40\ \Omega$ and an inductor with an inductance of $0.3/\pi\ \text{H}$.

(a) Calculate the frequency in the secondary coil.

(b) Determine the secondary voltage of the transformer.

(c) Calculate the rms voltage across each element (resistor and inductor).

(d) Calculate the efficiency of the transformer.

⑨ A proton is accelerated from an electric field into a magnetic field with an initial speed $v_0 = 10^3\ \text{km/s}$. The velocity vector is perpendicular to the magnetic induction. The motion of the proton in the magnetic field follows a circular path with a radius of $2\ \text{cm}$. Calculate the magnetic induction and the magnetic force acting on the proton. Given: mass of proton $1.67 \times 10^{-27}\ \text{kg}$ and charge $1.6 \times 10^{-19}\ \text{C}$.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination:** .....

Room N°: ..... **Seat N°:** ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

13

Questions and Exercises

① How can the direction of the induced current in a wire be determined using Lenz's 1st Law and Lenz's 2nd Law?

② What are the two strokes in a two-stroke gasoline engine?

③ State the right-hand rule for determining the direction of the induced current in a current-carrying wire of length l, moving with velocity v in a uniform magnetic field.

④ The equation of a transverse wave is given by: $y = A\sin(\omega t - kx + \phi)$ where parameter $A = 5\ \text{dm}$, $\omega = 50\pi\ \text{rad/s}$, $\phi = \pi/2\ \text{rad}$. Calculate the period, wavelength, and wave speed, and rewrite the wave equation.

⑤ A spherical balloon contains an ideal gas with a volume of $24.98\ \text{L}$ at a pressure of $1.5\ P_{atm}$ nd a temperature of $27\ °\text{C}$. The atmospheric pressure is constant at $P_{atm}=1\ \text{atm}$.

The universal gas constant is given as $R=8.31\ \text{J/mole. K}$, and the Avogadro constant is $N_A = 6.02\times10^{22}\ \text{/mole}$.

(a) Calculate the number of moles of the gas.

(b) Calculate the number of molecules contained in the balloon.

(c) Calculate the average translation kinetic energy of each molecule.

⑥ An engine has an average power of $3.344\ \text{hp}$ and a thermal efficiency of 0.44 for fuel combustion. One cycle takes $100\ \text{ms}$. (Take $1\ \text{hp} = 750\ \text{W}$.)

(a) Calculate the amount of heat absorbed by the engine in one cycle in order to do work.

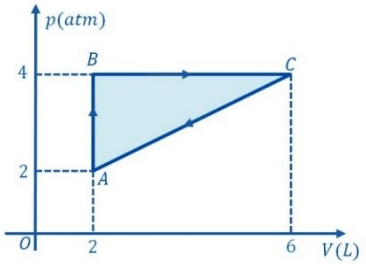
(b) Calculate the amount of heat expelled every cycle.

(c) In a duration of $2\ \text{hours}$, how much fuel is used by the engine? Given the fuel's latent $L = 4.03 \times 10^7\ \text{J/L}$.

⑦ A monoatomic ideal gas undergoes a thermodynamic process as shown in the diagram.

(a) Calculate the work done by the gas in one cycle.

(b) Calculate the amount of heat exchanged between the gas and the surroundings in one cycle.



⑧ After being fully charged, the capacitor has a maximum energy of $5 \times 10^{-5}\ \text{J}$, and the capacitance is $1\ \mu\text{F}$. Calculate the potential difference across the capacitor and the charge on the capacitor.

⑨ A $50\ \text{m}$ long wire is wrapped around a long cylinder with a length of $1\ \text{m}$ and a diameter D, forming a solenoid.

(a) What is the self-inductance of the solenoid?

(b) If the turns are not wrapped adjacently, what is the total number of turns if the wire has a diameter of $0.8\ \text{mm}$ and is covered by insulation with a thickness of $0.1\ \text{mm}$?

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

14

Questions and Exercises

① A spherical balloon of volume $4.00 \times 10^3\ \text{cm}^3$ contains helium at a pressure of $1.20 \times 10^5\ \text{Pa}$. How many moles of helium are in the balloon if the average kinetic energy of the helium atoms is $3.60 \times 10^{-22}\ \text{J}$?

② A gas molecule at the surface of earth happens to have the rms speed for the gas at $0\ °\text{C}$. Suppose it went straight up without colliding with other molecules, how high it would rise? Mass of the molecule is 4.65×10^{-26} kg and Boltzmann's constant $= 1.38 \times 10^{-23}\ \text{J/ K}$ and gravity $g = 9.8\ \text{m/s}^2$.

③ Calculate the work done during the thermodynamic process for the following cases:

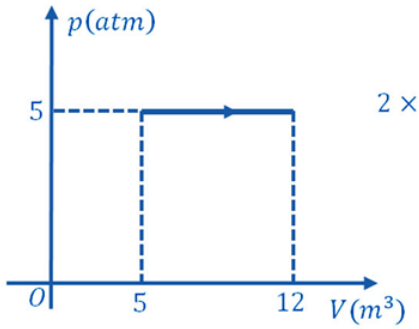


Fig A

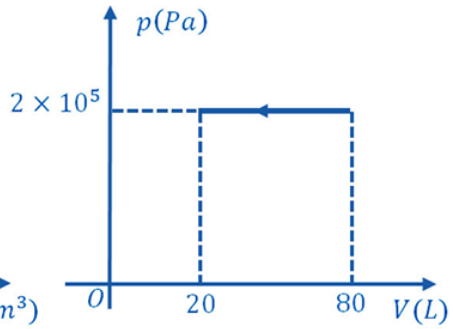


Fig B

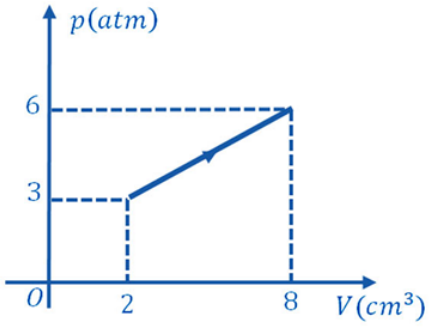


Fig C

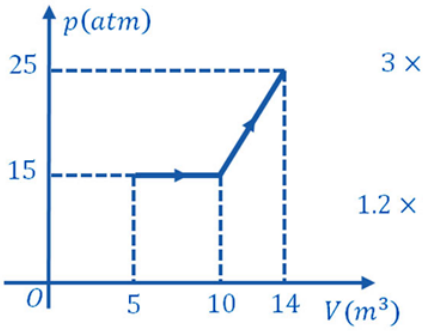


Fig D

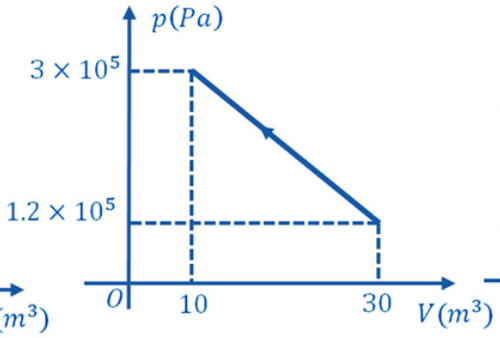


Fig E

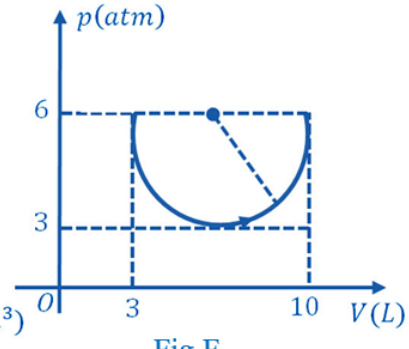


Fig F

④ A four-stroke gasoline engine does $450\ \text{J}$ of mechanical work every cycle with a thermal efficiency of 45%. The work done by the piston is transferred to the crankshaft with a mechanical efficiency of 90%.

(a) Calculate the amount of heat received from the combustion of fuel to do work.
(b) Calculate the mechanical work transferred to the crankshaft.
(c) Calculate the amount of heat expelled due to friction in the gears of the engine.
(d) Calculate the power of the engine during this operation over $1\ \text{hour}$.

⑤ Two oscillators in a medium have the following equations:

$$y_1 = (\sqrt{2}\ \text{dm})\sin\left(10\pi t + \frac{\pi}{4}\right) \text{ and } y_2 = (\sqrt{2}\ \text{dm})\sin\left(10\pi t + \frac{3\pi}{4}\right)$$

(a) Calculate the frequency of each oscillator.
(b) Calculate the resultant amplitude and phase constant of the oscillators.

⑥ A series circuit consists of a capacitor and an inductor connected to an AC source with a frequency of $50\ \text{Hz}$ and an rms voltage of $40\ \text{V}$. The capacitor has a capacitance of $10^{-3}/3\pi\ \text{F}$, and the inductor has a self-inductance of $4\pi \times 10^{-2}\ \text{H}$.

(a) What is the rms current running through the circuit?
(b) What is the apparent power in the circuit?
(c) What is the rms voltage across the terminals of each component?

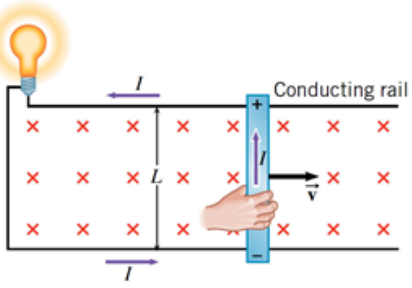
⑦ An electrical wire with a length of $314\ \text{m}$ is wrapped around a cylinder with a diameter of $10\ \text{cm}$, forming a solenoid with a length $l = 1\ \text{m}$.

(a) What is the diameter of the wire, assuming the thickness of the insulation is negligible?
(b) If an electric current of $2/\pi\ \text{A}$ is running through the solenoid, determine the magnetic field at the center of the solenoid. Given: $\mu_0 = 4\pi \times 10^{-7}\ \text{Tm/A}$.

⑧ A 100 cm solenoid with a diameter of $10\ \text{cm}$ is formed by a wire with a diameter of 1 mm, neglecting the thickness of insulation and wrapped closely.

(a) What is the resistance of the solenoid? Given that the resistivity of the wire is $\rho = 1.6\ \mu\Omega\ \text{cm}$.
(b) What is the self-inductance of the solenoid?
(c) The current varies as $i = t^2 - 2$, where i is measured in amperes and t in seconds. What is the voltage across the terminals of the solenoid when $t = 1\ \text{s}$, $\pi^2 = 10$.

⑨ Suppose that the rod in Figure is moving at a speed of $5.0\ \text{m/s}$ in a direction perpendicular to a 0.80-T magnetic field. The rod has a length of $1.6\ \text{m}$ and a negligible electrical resistance. The rails also have negligible resistance. The light bulb, however, has a resistance of $96\ \Omega$. Find



(a) the emf produced by the rod,
(b) the current induced in the circuit,
(c) the electric power delivered to the bulb, and
(d) the energy used by the bulb in $60.0\ \text{s}$.

⑩ When a long iron-core solenoid is connected across a 6-V battery, the current rises to its maximum value after a time of $0.75\ \text{s}$. The experiment is then repeated with the iron core removed. Now the time required to reach 0.63 of the maximum is $0.0025\ \text{s}$. Calculate

(a) the relative permeability of the iron and
(b) the inductance L for the air-core solenoid if the maximum current is $0.5\ \text{A}$.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination:** .....

Room N°: ..... **Seat N°:** ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

15

Questions and Exercises

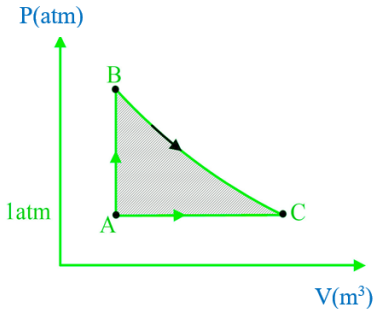
① Determine the root-mean-square speed (v_{rms}) of a helium atom at the surface of the Sun, where the temperature is approximately 6000 K. Assuming the mass molecule of proton is $m_0 = 1.67 \times 10^{-27}$ kg.

② 500 J of heat is added to a monoatomic nitrogen gas in an isochoric process at low temperature. The gas is helium with a mass of 4 g. The universal gas constant is $R = 8.31$ J/mol. K, and the molar mass of helium is $M_{He} = 4$ kg/kmol.

(a) What is the increase in temperature of the gas in degrees Celsius?

(b) Does this gas do any work? If so, explain why or why not.

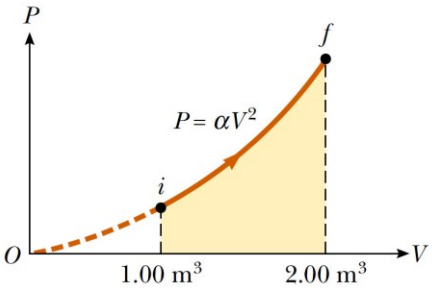
③ 1.00 kmole of helium gas undergoes the engine operation as shown in the diagram below. The process BC is isothermal, with $P_A = 1$ atm, $V_A = 22.4$ m^3, and $P_B = 2$ atm. Assuming helium behaves as an ideal gas.



(a) What are the temperatures T_A, T_B, and the volume V_C?

(b) How much work is done by the surroundings?

④ An ideal gas is taken through a quasi-static process described by $P = \alpha V^2$, with a $\alpha = 5.00$ atm/m^6, as shown in Figure. The gas is expanded to twice its original volume of 1.00 m^3. How much work is done on the expanding gas in this process?



⑤ An ideal engine operates between reservoirs at temperatures T_C and $T_H = 227$ °C, absorbing 300 kJ of heat from the hot reservoir at high temperature. The engine expels 200 kJ of heat to the surroundings.

(a) What is the temperature T_C?

(b) What is the efficiency of the ideal engine?

(c) How much work is done in one cycle?

⑥ Find the resultant wave of these three oscillations below.

$y_1 = (7\text{cm})\cos(5\pi t)$

$y_2 = (4\text{cm})\sin(5\pi t - \pi/2)$

$y_3 = (4\text{cm})\sin(5\pi t)$.

Given $\tan^{-1}(0.75) = 0.64$ rad.

⑦ An inductor with an inductance of $20\ \text{mH}$ is connected to an AC voltage with the instantaneous voltage expression: $V = (20\sqrt{2}\ \text{V})\sin\left(50t + \dfrac{5\pi}{6}\right)$

(a) What is the self-inductance of the inductor?

(b) Write the instantaneous current through the inductor.

⑧ A $50\ \text{m}$ long wire is placed between two poles of a magnet, where the magnetic field between the poles is uniform with a magnitude of $10\ \text{mT}$. How fast should the wire be moved (in m/s) in order to produce an induced emf of $0.1\ \text{V}$? The wire moves perpendicular to the magnetic field.

⑨ A 1000-turn coil carries a current of $10\ \text{A}$ and has a length of $80\ \text{dm}$.

(a) What is the magnetic field inside the coil?

(b) What is the self-inductance of the coil if the radius of the coil is $2\ \text{cm}$?

(c) What is the induced emf if the current decreases at a rate of $0.2\ \text{A/s}$?

(d) What is the magnetic energy stored in the coil?

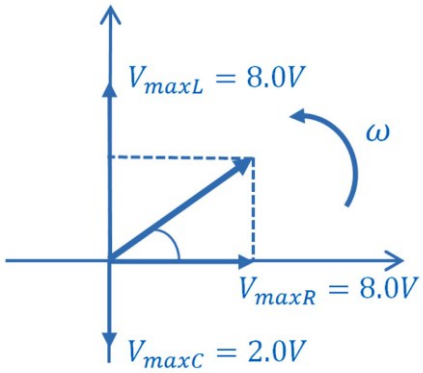
(e) What is the magnetic flux due to self-induction in the coil?

⑩ The voltage diagram of a resonance RLC circuit is shown in the Figure. Given $R = 50\ \Omega$ and a source frequency of $60\ \text{Hz}$, find:

(a) The maximum voltage V_{max}

(b) The maximum current I_{max}

(c) The capacitance, inductance, and average power.



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

16

Questions and Exercises

① What is a standing wave, and what are the nodes and antinodes of a wave?

② Write the first law of thermodynamic.

③ Nuclear fusion occurs among deuterium nuclei when the average kinetic energy of a nucleus is 0.72 MeV. At what temperature can this nuclear fusion occur? $1\text{ev} = 1.6\times10^{-19}\text{J}$, $k_B = 1.38\times10^{-23}$ J/K.

④ In a period of 1.00 s, 5.00×10^{23} nitrogen molecules strike a wall with an area of 8.00 cm^2. Assume the molecules move with a speed of 300 m/s and strike the wall head-on in elastic collisions. What is the pressure exerted on the wall? Note: The mass of one N_2 molecule is 4.65×10^{-26} kg.

⑤ A container holds 1 mole of an ideal monoatomic gas at an initial temperature of $27°C$. While maintaining a constant temperature, the gas expands from an initial volume from $V_1 = 340$ dm^3 to V_2. Give: $R = 8.31$ J/mol.K.

(a)What is the internal energy of the gas?

(b) If the work done by the gas during expansion is 1700 J, how much heat is absorbed by the gas?

(c) What is the final volume V_2 of the gas in liters?

⑥ An isothermal expansion of an ideal gas occurs at an initial pressure of 3040 mmHg, allowing the volume to increase from 2 L to 6 L.

(a) Calculate the work done by the gas during this process.

(b) Calculate the temperature of the gas if it has 0.5 moles. Given that $R = 8.31$ J/mol.K, $\ln(6) = 1.80$, $\ln(2) = 0.70$, and $\ln(3) = 1.10$.

⑦ Two oscillators with the same frequency of 50 Hz exist in the same medium, where each oscillator has a given amplitude and phase constant: $a_1 = 0.1$ m, $a_2 = 0.2$ m, $\phi_1 = \pi/2$ rad, $\phi_2 = \pi$ rad.

(a) Calculate the period and frequency of each oscillator.

(b) Write the equation of motion for each oscillator as a function of time.

(c) Determine the resultant equation of the oscillation. Given: $\tan^{-1}(0.5) = 0.64$ rad.

⑧ An infinitely long straight wire carries a current of 10 A in air. A point M is located 5 cm from the wire. Take permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ Tm/A.

(a) Calculate the magnetic field produced by the current-carrying wire at point M.

(b) If the magnetic field at point N has a magnitude of 0.5×10^{-5} T, calculate the distance from the wire to point N.

⑨ A flat coil with an average radius R and 50 turns is forcefully moved into a uniform magnetic field. The rate of change of magnetic flux through the coil is 0.1 Wb/s.

(a) Calculate the induced emf produced by the coil.

(b) After the above condition, the coil is quickly pulled away, and the induced emf is measured to be 9V. What is the required rate of change in magnetic flux for the coil?

⑩ The useful work output of an actual engine is only 30% of the heat energy supplied from the combustion of fuel. Additionally, 96% of the piston's work is transferred to the crankshaft, as a large portion of heat is expelled into the atmosphere, and a small portion is lost due to friction in the engine's gears.

(a) Calculate the combustion efficiency of the engine.

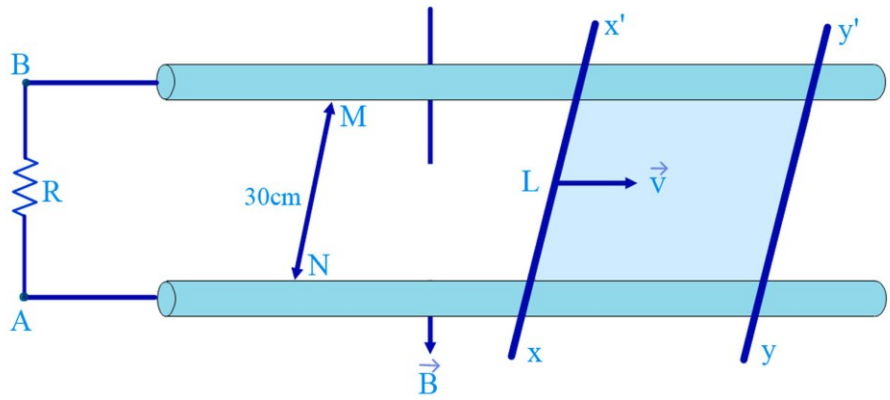
(b) Determine the work done by the crankshaft if the useful work output is 5400 J.

⑪ As shown in the figure: Initially, a conducting bar is positioned at xx' and is then pulled with a force of 10 N, moving until it reaches yy'. The length of xx' is 30 cm, and the motion takes 0.5 s. The magnetic field strength is $B = 400$ mT.

(a) Calculate the induced emf in the bar.

(b) Determine the direction of the induced current through the resistor R, and calculate the magnitude of the induced current if $R = 0.72$ Ω.

(c) Calculate the electrical power dissipated in the resistor and the average power exerted by the force moving the bar.

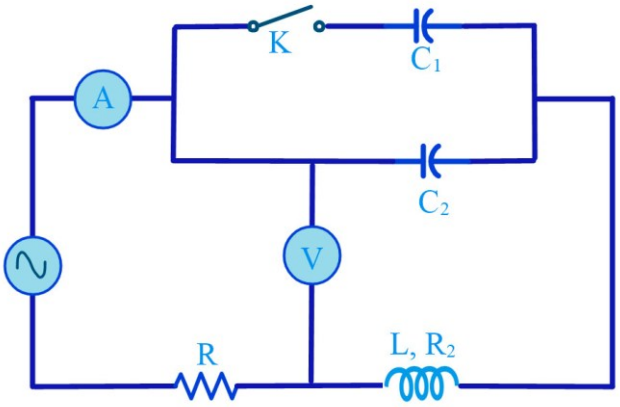


⑫ An AC voltage source with an RMS voltage of 200 V and an angular frequency of 300 rad/s is connected to a circuit. The component values are as follows: $C_1 = C_2 = \frac{100}{12}\,\mu\text{F}, L = \frac{1}{3}\,\text{H}, R_1 = R_2 = 30\,\Omega$ and $\sqrt{73} = 8.54$

When the switch is closed:

(a) Determine the reading on the ammeter.

(b) Determine the reading on the voltmeter.



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

17

Questions and Exercises

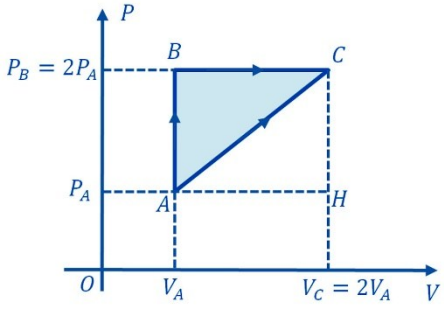
① What are the sources of magnetic field? What is the SI unit of magnetic induction?

② State the principle of initial and final state.

③ (a) Show that the density of an ideal gas occupying a volume V is given by, $\rho = \dfrac{PM}{RT}$, where M is the molar mass.

(b) Determine the density of oxygen gas at atmospheric pressure and $20.0°\text{C}$. Given the molar mass of oxygen $(M_{O2}) = 32\ \text{g/mole}$.

④ A monoatomic gas consisting of n kmol undergoes a quasi-static process from state A to state C along the given path in the diagram.



(a) Determine the work done by the gas, the change in internal energy, and the heat supplied to the gas as functions of P_A and V_A during this process.

(b) Determine the work done by the gas, the change in internal energy, and the heat supplied to the gas as functions of P_A and V_A if the gas undergoes a quasi-static process from state A to C along path ABC.

(c) Identify two similarities and two differences between the results obtained in parts (a) and (b).

⑤ A Carnot engine operates between a cold reservoir at $20°\text{C}$ with an initial thermal efficiency of 60%. When the temperature of the hot reservoir is increased, the efficiency rises to 75%. Calculate the required increase in the hot reservoir's temperature in degrees Celsius.

⑥ A standing wave is described by the equation: $y_{(x,t)} = (4\text{cm})\ \sin\ (30.5\ \text{rad/m} \cdot x)\ \cos\ (680\ \text{rad/s} \cdot t)$. For two waves formed a standing wave. Calculate the amplitude, wavelength, frequency, and wave speed.

⑦ An inductor is connected to a DC voltage source of $48\ \text{V}$, resulting in a current of $8\ \text{A}$ flowing through it. When the inductor is connected to an AC voltage source with a frequency of $50\ \text{Hz}$ and an rms voltage of $120\ \text{V}$, the rms current is measured to be $12\ \text{A}$. Determine the following:

(a) The electrical resistance of the inductor.

(b) The inductance of the inductor.

⑧ An undamped oscillation in an LC circuit is described by the charge equation: $q_{(t)} = (100\ \mu\text{C})\ \cos(1000t)$. Determine the following:

(a) The instantaneous current as a function of time.

(b) The capacitance of the capacitor (in microfarads) given that the voltage across it is $10\ \text{V}$.

(c) The inductance of the inductor (in millihenries).

⑨ A square coil of wire with an edge length of $50\ \text{cm}$ is placed in a uniform magnetic field of magnitude $0.16\ \text{T}$. Calculate the magnetic flux through the surface of the coil in the following cases:

(a) When the magnetic field is perpendicular to the surface of the coil.

(b) When the magnetic field is at an angle of $30°$ to the surface of the coil.

(c) Determine the magnitude of the average induced current in the coil if the wire has a resistance of $0.012\ \Omega$ and is moved from the orientation in part (b) to (a) over a time interval of $0.14\ \text{s}$.

⑩ An RL circuit with a resistance $R = 10\ \Omega$ and an inductance $L = 0.1\ \text{H}$ is connected in series to a DC source with an emf of $12\ \text{V}$. Given that $1/e^2 \approx 0.082$, and at time $t = 0$, the switch is closed.

(a) What is the time constant of the circuit?

(b) What is the instantaneous current in the circuit at $t = 0.025\text{s}$?

(c) What is the steady-state current in the circuit after the switch has been closed for a long time $(t \to \infty)$.

⑪ A series RLC circuit consists of: Resistor $R_1 = 100\ \Omega$, inductor with an inductance of $L = 0.314\ \text{H}$ and a resistance of $R_2 = 100\ \Omega$ and capacitor with a capacitance of $C = 0.314 \times 10^{-4}\ \text{F}$. This arrangement is connected to an AC voltage source with the voltage expression: $v = (200\sqrt{2}\ \text{V})\sin\left(100\pi t\right)$

(a) Determine whether the rms current in the circuit reaches its maximum value. Given: $\pi^2 = 10$.

(b) Find the voltage expression across the terminals of each component (resistor, inductor, and capacitor).

⑫ A solenoid has a resistance of $30\ \Omega$ and inductance L, and it carries a varying current i. This solenoid is tightly wound over a length of $500\ \text{mm}$ with a radius of each turn being $50\ \text{mm}$. The diameter of the wire used to wrap the solenoid is $0.5\ \text{mm}$.

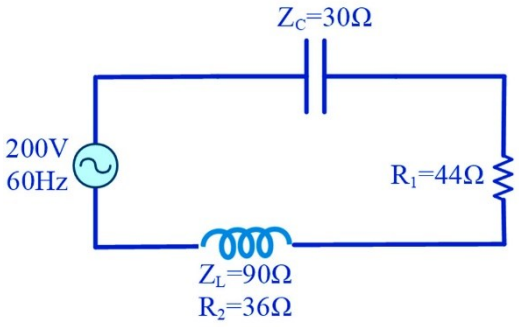
(a) Calculate the inductance of the solenoid. Given: $\pi^2 = 10$.

(b) Write the expression for the voltage across the terminals of the solenoid when the varying current is given by the equation: $i\ (\text{t}) = t^2 - 5\ (\text{A})$.

⑬ A circuit consists of an AC voltage source, a capacitor, a resistor, and an inductor (with both inductance and resistance), all connected in series with each other.

(a) Calculate the rms current flowing through the circuit.

(b) Calculate the rms voltage across the terminals of the capacitor and the inductor. Given: $\sqrt{9396}$



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination:** .....

Room N°: ..... **Seat N°:** ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

18

Questions and Exercises

① For waves, why are Fresnel equations constructed?

② What is an adiabatic process? What is the formular of this process?

③ Assuming 1.80 moles of ideal gas are transferred from a container with a volume of $0.3\ \text{m}^3$ to another container with a volume of $1.5\ \text{m}^3$ through an isothermal compression at a temperature of $30\ ^\circ\text{C}$.

(a) Calculate the energy transformed to heat during compression.

(b) Does this energy get transferred to the gas or by the gas? Given $R = 8.31\ \text{J/mol. K}$ and $\ln(0.5) = -0.7$

④ 2 moles of oxygen are contained in a 5 L container under a pressure of 8 atm. Calculate the average kinetic energy of each oxygen molecule in these conditions.

⑤ (a) A thermodynamic system absorbs 200 cal of heat and does 120 cal of work. Does the change in internal energy increase or decrease, and what is the change in internal energy in Joules?

(b) A thermodynamic system releases 4500 kJ of heat, and the change in internal energy decreases by 2400 kJ. Does the system do work or have work done on it by the surroundings, and what is the work in Joules?

⑥ A 1-mole monoatomic ideal gas undergoes a process at constant atmospheric pressure ($P_{\text{atm}} = 10^5\ \text{Pa}$) until its volume decreases to 50% of its initial volume. During this process, 200 J of work is done on the gas. Given the universal gas constant $R = 8.31\ \text{J/mole. K}$.

(a) Calculate the initial and final volumes of the gas.

(b) Calculate the change in temperature of the gas.

⑦ Two oscillators produce two oscillations respectively. The first oscillation has an amplitude of 5 cm with no phase constant, while the second oscillation has an amplitude twice that of the first oscillation and lags by a phase angle of $\pi/2$ rad. Both oscillations oscillate with the same period of 50 ms.

(a) Write the equation for each oscillation.

(b) Write the resultant equation of the combined oscillation. Given: $\tan^{-1}(2) = 63.43°$

⑧ Two identical waves propagate in opposite directions through a medium, each with the respective equations:

$$y_1 = (3\,\text{cm})\sin\left(10\pi t - \frac{\pi}{2}x\right) \text{ and } y_2 = (3\,\text{cm})\sin\left(10\pi t + \frac{\pi}{2}x\right)$$

(a) Write the resultant wave. What is the nature of this resultant wave?

(b) Calculate the amplitude at distance $x = 8$ cm. What is the position of the wave?

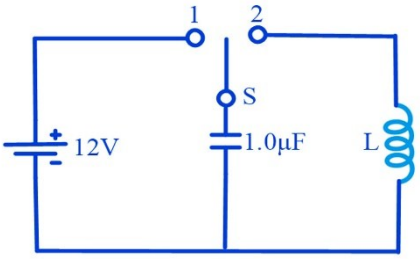
(c) Calculate the position of maximum amplitude and minimum amplitude.

⑨ A copper wire with a diameter of $1.0\ \text{mm}$, neglecting insulation, is wound around a cylinder with a length of 1.0 m and a diameter of $4.0\ \text{dm}$ to form a solenoid.

(a) How long is the copper wire used?

(b) Find the resistance of the solenoid if the resistivity of copper is $\rho_{Cu} = 1.72\ \mu\Omega\text{cm}$.

⑩ The circuit, as shown in the figure, operates as follows: The switch is first moved to position 1, causing the capacitor to charge. Afterward, the switch is moved to position 2, and the capacitor discharges its electric charge to the inductor, creating an electric oscillation. This circuit oscillates with an angular frequency of $1000\ \text{rad/s}$. Assuming the oscillation is undamped:



(a) Calculate the inductance of the inductor.

(b) Write the charge equation of the capacitor.

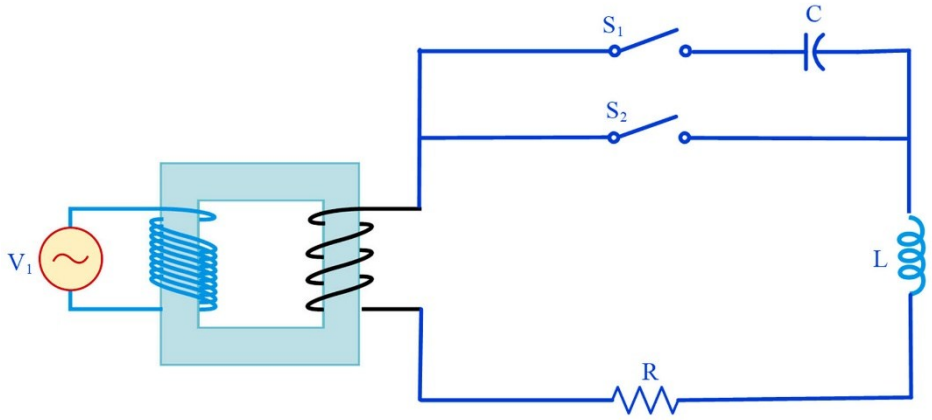
(c) Write the equation for the instantaneous current flowing through the circuit.

⑪ The circuit, as shown in the figure, a step-up transformer with a primary voltage of $50\ \text{V}$, angular frequency of 300 rad/s and 500 turns, and a secondary coil with $1000\ \text{turns}$, a capacitor with a capacitance of $10^{-3}/18\ \text{F}$, a negligible internal resistance in the inductor with an inductance of $1/3\ \text{H}$, and a resistor with a resistance of $30\ \Omega$.

(a) Find the value of the current flowing through the circuit when switch S_1 is closed and switch S_2 is open.

(b) Find the value of the current flowing through the circuit when switch S_1 is open and switch S_2 is closed.

Given: $\sqrt{10900} = 104.4$



⑫ An alternator has a power factor of 0.9 and provides an rms current of $200\ \text{A}$ under a voltage of $220\ \text{V}$. The power lost by the alternator is $5.9\ \text{kW}$. Calculate the efficiency of the alternator.

⑬ A solenoid has $2000\ \text{turns per meter}$, with a volume of $\pi\ \text{m}^3$ and carries a current described by $i\ (t) = 20 - 2t^2\ (\text{A})$. Calculate the induced emf at an instant $t = 0.2\ \text{s}$. Given: $\mu_0 = 4\pi \times 10^{-7}\ \text{Tm/A}$.

⑭ An electron is moving with a velocity of $0.2c$ (where c is the speed of light) into a region with a magnetic field of magnitude $4000\ \mu\text{T}$. The angle between the velocity of the electron and the magnetic field is $30°$. Calculate the magnetic force acting on the electron. Given that $c = 3 \times 10^8\ \text{m/s}$, the charge of the electron $q = -1.6 \times 10^{-19}\ \text{C}$, $\sin\ (30°) = 0.5$, and $\cos\ (30°)$.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

19

Questions and Exercises

① What is an ideal engine?

② What is a solenoid?

③ What is a four-stroke gasoline engine? Among the 4 processes, which process does the gas do work?

④ A 0.1 mole of a monoatomic ideal gas undergoes a thermodynamic process at a constant temperature of 300 K. The gas expands from an initial volume $V_i = 1\ \text{dm}^3$ to a final volume V_f, and does 174.51 J of work during the expansion.

(a) Sketch a PV diagram that represents this change in volume.

(b) Calculate the final volume of the gas after the expansion.

(c) Determine the amount of heat supplied to the gas during this process. Given: $R = 8.31\ \text{J/mol. K}$, $\ln 2 = 0.7$

⑤ The temperature of 2 moles of an ideal monatomic gas decreases from an initial temperature $T_i = 540$ K to a final temperature $T_f = 350$ K. During this process, 5500 J of heat is released to the surroundings.

(a) Calculate the change in internal energy of the gas.

(b) Calculate the work done by the gas.

⑥ A four-stroke engine delivers an output power of 5.0 hp through its four processes and operates with a thermal efficiency of 32 %.

(a) Identify the process in which the engine performs work.

(b) Calculate the amount of heat absorbed by the engine in one cycle, given that one cycle takes 0.20 s.

(c) Determine the amount of fuel consumed by the engine in one hour, assuming $1\ \text{hp} = 746$ W and the fuel's latent heat $L = 4.03 \times 10^7\ \text{J/L}$.

⑦ A car engine receives 150 kJ of heat from gasoline-air mixture every second and expels 120 kJ of heat to the atmosphere.

(a) What is the mechanical work done during 5 minutes?

(b) What is the thermal efficiency of the engine?

(c) If the mechanical efficiency is 92%, how much work is done and useful efficiency by the crankshaft?

⑧ A solenoid has a length much greater than its radius, and it has a metal core placed along its length. A current of 1.0 A passes through the solenoid, and a Tesla meter measures the magnetic field value as 1.256×10^{-2} T. The number of turns per meter is $1000\ \text{turns/m}$. Find the permeability of the iron core. Given: $\mu_0 = 4\pi \times 10^{-7}\ \text{Tm/A}$.

⑨ Construct a Fresnel diagram to determine the resultant equation of two oscillations:

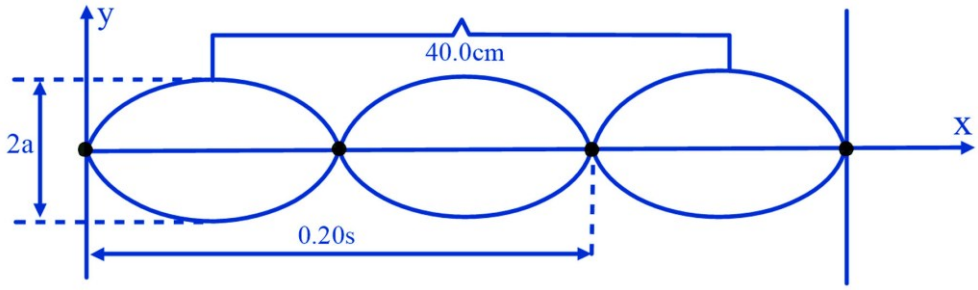
$y_1 = (6\text{cm})\cos(2\pi ft)$, and $y_2 = (6\text{cm})\sin(2\pi ft)$.

⑩ As illustrated in the figure below:

(a) Determine the number of node and antinode positions in the resulting standing wave.

(b) Calculate the wave speed.

(c) Write the wave equation that describes the resultant wave.

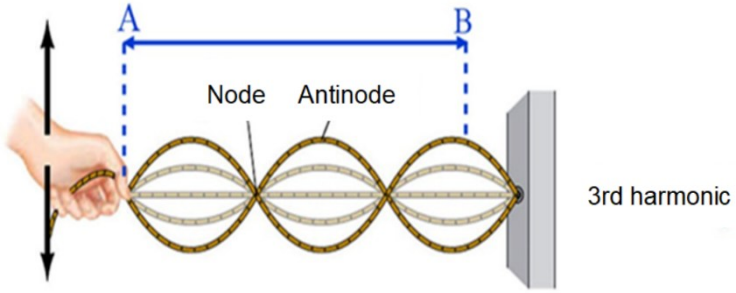


⑪ A rope is shaken with a fixed frequency, producing a wave with the equation: $y = (6\text{ cm})\sin(1/5\pi x)\cos(10\pi t)$ as shown in Fig.

(a) Determine the position of nodes and antinodes.

(b) Prove that the distance from point A to point B is 12.5 cm.

(c) Calculate the wave propagation speed in meters per second (m/s).



⑫ A solenoid with a volume of $V = 0.314\text{ m}^3$ and a length of $l = 1\text{ m}$ is wound with a high-quality insulated wire into closely packed turns N. The solenoid has a self-inductance of $L = 100\text{ mH}$.

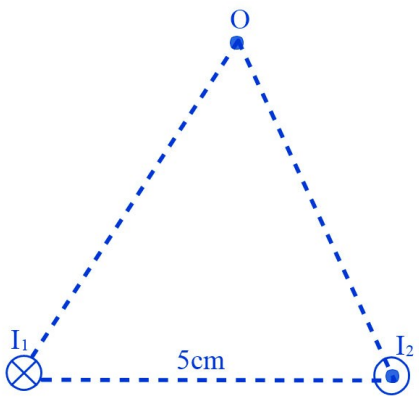
(a) Calculate the diameter of the solenoid.

(b) A time-varying current $i = 2t^2 + 3\text{ (A)}$ passes through the solenoid. Write the expression for the induced electromotive force (emf) in the solenoid.

(c) Calculate the total number of turns N in the solenoid. Given: Given: $\mu_0 = 4\pi\times10^{-7}\text{H/m}$ and $\pi^2 = 10$.

⑬ Two long wires carry currents I_1 and I_2 with the same magnitude $I_1 = I_2 = 10\text{ A}$, as shown in the figure. The directions of the currents are indicated in the figure. Determine the magnetic field at point O.

Given: $\mu_0 = 4\pi\times10^{-7}\text{ Tm/A}$.



Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

20

Questions and Exercises

① Determine the volume occupied by 3.2 grams of oxygen gas in a container at a pressure of 1.0×10^5 Pa and a temperature of 27 °C. The universal gas constant is given as $R = 8.31$ J/mole. K.

② A system undergoes a process where 20 kJ of work is done on it. Subsequently, 1 kcal of heat is released from the system. Calculate the change in internal energy of the system. Given: 1cal=4.186J.

③ Calculate the average translational kinetic energy of a gas molecule at a temperature of 1727 °C. Given: Universal gas constant $R = 8.31$ J/mol.K and Avogadro's number $N_A = 6.00 \times 10^{23}$ molecules/mol.

④ A car engine has an efficiency of 0.40 and absorbs 5.0 MJ of energy. Calculate:

(a) The mechanical work produced by the piston.
(b) The amount of heat expelled to the atmosphere.
(c) The useful work if the mechanical efficiency is 0.80.

⑤ Calculate the change in internal energy of the system:

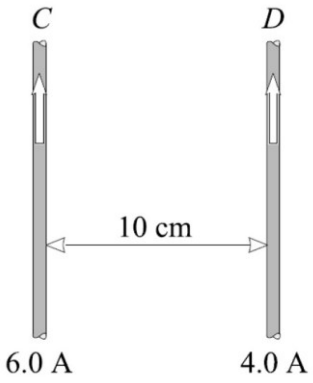
(a) The system does 500 J of work during an adiabatic process.

(b) During adiabatic compression, 1000 J of work is done on the gas.

⑥ The displacement of a wave is described by the equation: $y = 0.30 \sin(0.20\,x - 0.20t)$ (m). Calculate:

(a) The amplitude of the wave.
(b) The wave number.
(c) The wavelength.
(d) The period of the wave.
(e) The wave speed.

⑦ Two long parallel straight wires are separated by a distance of 10 cm. One wire carries a current of 6.0 A and the other carries a current of 4.0 A. Calculate the magnetic force acting on a 1-meter length of wire (wire D) in the following two cases:



(a) When the currents in both wires are in the same direction.
(b) When the currents in both wires are in the opposite direction.

⑧ A solenoid with 470 turns per unit length has a total length of 1.5 m and stores 0.31 J of magnetic energy when a current of 12 A flows through it. Given $\mu_0 = 4 \times 10^{-7}$ T·m/A.

(a) Calculate the inductance L of the solenoid.

(b) Calculate the cross-sectional area A of the solenoid.

⑨ A particle with an electric charge of $q = 1.6\times10^{-19}$ C moves with a velocity of $v = 5.0\times10^{5}$ m/s in a circular path of radius $r = 2.0\times10^{-2}$m when it enters a uniform magnetic field. The direction of the particle's motion is perpendicular to the magnetic field, which has a magnitude of $B = 0.26$T. Calculate the mass of the particle.

⑩ Two cylindrical rods are placed parallel to each other in a horizontal plane, with their ends connected to a resistance of $R = 6.0\ \Omega$. The rods are separated by a distance of $d = 30$ cm. A metal bar MN is placed perpendicular to the two rods. The setup is located in a uniform magnetic field of magnitude $B = 0.20$ T, directed perpendicular to the plane of the rods. The metal bar MN is moved along the rods at a constant speed of $v = 100$ m/s. Assuming the resistance of the metal bar and the rods is negligible, calculate the magnitude of the induced electromotive force (emf) across the resistance.

⑪ A solenoid of length $l = 1.0$ m with $N = 500$ turns stores a magnetic energy of $0.5\ \pi$ J when a varying current of $I = 5$ A passes through it. Given the permeability of free space $\mu_0 = 4\pi\times10^{-7}$ Tm/A.

(a) Calculate the inductance of the solenoid.

(b) Calculate the cross-sectional area of the solenoid.

⑫ An air-core solenoid has a length of $l = 50$ cm and a diameter of 3.0 cm, wound with $N = 3000$ turns. If the solenoid carries a current of $I = 5.0$ A, calculate:

(a) The magnetic field inside the solenoid.

(b) The length of the wire used to wind the solenoid. Given the permeability of free space $\mu_0 = 4\pi\times10^{-7}$ Tm/A.

⑬ A copper wire with a cross-sectional area of $A = 0.2$ mm^2 and resistivity $\rho = 1.7\times10^{-8}\ \Omega$m is wound into an air-core solenoid with $N = 6000$ turns. The solenoid carries a current of $I = 1.0$ A. Given the permeability of free space $\mu_0 = 4\pi\times10^{-7}$ Tm/A. Calculate:

(a) The magnetic field inside the solenoid.

(b) The length of the wire used to wind the solenoid.

(c) The resistance of the wire.

⑭ An inductor with an inductance of $L = 64$ mH is constructed using a wire of diameter $d = 0.8$ mm, covered by insulation with a thickness of $e = 0.1$ mm. The wire is wound into tightly packed turns on a cylinder with a diameter of $D = 2$ cm.

(a) Calculate the number of turns required to construct the coil.

(b) If the inductance is to be increased to $L_2 = 81$ mH by rewinding the coil using a thinner wire on the same cylinder, determine the number of turns needed for the new coil. Given: $\pi^2 = 10$.

⑮ A transformer consists of two coils: one with $N_1 = 10000$ turns and another with $N_2 = 200$ turns. Which coil should be used as the primary coil to obtain a voltage of 220 V at the secondary coil?

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination:** .....

Room N°: ..... **Seat N°:** ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ **No materials, including books, documents, or cell phones, are permitted during exams.**

▪ **You're not allowed to cheat in an exam.**

▪ **A calculator is not permitted during an exam.**

21

Questions and Exercises

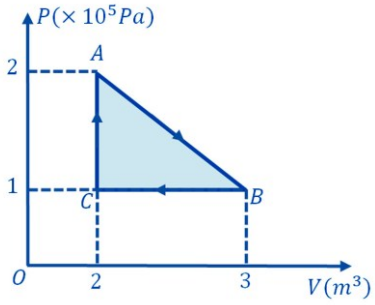
① When current passes through a coil, it is observed that the magnetic field lines emerge from one pole and enter the other pole. Which pole is the north pole, and which pole is the south pole of the coil?

② Calculate the volume occupied by 6.4 grams of oxygen gas at a pressure of 10^5 Pa and a temperature of 400 K. The molar mass of oxygen is given as $M = 32$ g/mole.

③ Calculate the volume occupied by 2.8 grams of nitrogen gas under a pressure of 1.0×10^5 Pa and a temperature of 300 K. The universal gas constant is given as $R = 8.31$ J/mole. K, and the molar mass of nitrogen gas is $M = 28$ g/mole.

④ A capacitor with a capacitance of $C = 2.0\mu$F is charged to a voltage of $V = 5.0$V. Calculate the electric energy stored in the capacitor.

⑤ A thermodynamic process is performed as shown in the figure below. Calculate:



(a) The work done in the thermodynamic process from A to B.

(b) The work done in the thermodynamic process from B to C.

(c) The work done in the thermodynamic process from C to A.

(d) The net work done in the cyclic process $ABCA$.

⑥ Calculate the change in internal energy for each of the following scenarios:

(a) A system absorbs 2000 J of heat and does 500 J of work.

(b) A system absorbs 1200 J of heat and received 400 J of work.

(c) 300 J of heat is released from the system during a process at constant volume.

⑦ An engine has a thermal efficiency of 40%. Calculate:

(a) The work done by the engine if it absorbs 2000 J of heat from the hot reservoir.

(b) The heat expelled by the engine to the cold reservoir.

⑧ A diesel engine in a car has a thermal efficiency of 0.45 and absorbs 4.0×10^6 J of energy. Calculate:

(a) The mechanical work produced by the piston.

(b) The amount of heat expelled to the atmosphere.

(c) The useful work if the mechanical efficiency is 0.80.

⑨ A gasoline engine receives 4.0×10^6 J of energy and has a thermal efficiency of 0.40.

(a) What is the mechanical work produced by the piston?

(b) How much heat is expelled to the atmosphere?

(c) If the mechanical efficiency is 0.85, calculate the work done by the crankshaft.

⑩ In each of the following situations, find the change in internal energy of the system.

(a) A system absorbs $500\ \text{cal}$ of heat and at the same time does $400\ \text{J}$ of work.

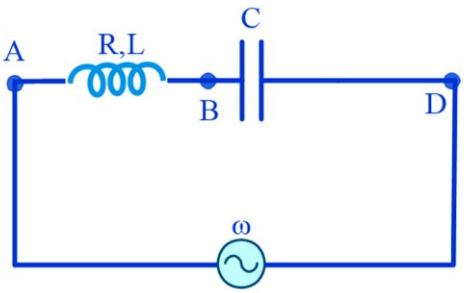
(b) A system absorbs $300\ \text{cal}$ and at the same time $420\ \text{J}$ of work is done on it.

(c) Twelve hundred calories is removed from a gas at constant volume. Give you answers in kilojoules.

⑪ An alternating current (AC) circuit with a given angular frequency is shown in the figure. The root-mean-square (rms) voltages are $V_{\text{AB}} = 160\ \text{V},\ V_{\text{BD}} = 56\ \text{V},\ V_{\text{AD}} = 120\text{V}$ and rms current $I = 0.2\text{A}$.

(a) Calculate the impedances $Z_{\text{AB}},\ Z_{\text{BD}}$, and Z_{AD}.

(b) Calculate the resistance of the coil.



⑫ Two wires of equal length $l_1 = l_2 = 1\ \text{m}$ are placed parallel to each other, carrying currents $I_1 = I_2 = 1\ \text{A}$ in the same direction. Given the permeability of free space $\mu_0 = 4\pi\times10^{-7}\ \text{Tm/A}$.

(a) Calculate the interaction force between the two wires.

(b) Do the wires attract or repel each other?

⑬ In a laboratory, an electron is accelerated from rest across a potential difference of $350\ \text{V}$. It then moves into a uniform magnetic field where its velocity is perpendicular to the magnetic field. The electron moves along a curved path due to the magnetic force acting on it, with a radius of $7.5\ \text{cm}$.

(a) Calculate the magnitude of the magnetic field.

(b) Calculate the angular velocity of the electron.

⑭ A solenoid with $100\ \text{turns}$, each loop having a cross-sectional area of $A = 1\text{dm}^2$, is placed in a uniform magnetic field with magnetic induction B=10^{-3}T. The normal to the plane of the loops makes an angle of $60°$ with the magnetic field. Calculate the magnetic induction at the center of the solenoid.

⑮ A current $I = 0.5\text{A}$ flows through an infinite wire placed in air.

(a) Calculate the magnetic induction at point M, located at a distance of $2\ \text{cm}$ from the wire.

(b) Calculate the magnetic induction at point M, where point M is located in a medium with a relative permeability of $\mu_r = 1000$.

⑯ An inductor with resistance $R = 1\ \Omega$ and inductance $L = 1\ \text{H}$ is connected in series in an AC circuit with a frequency of $f = 50\ \text{Hz}$ and an rms voltage of $V = 115\text{V}$. Calculate:

(a) The capacitance of the capacitor required to achieve resonance in the circuit.

(b) The rms current at resonance. Given $\pi^2 = 10$.

Exam Center: Western International School | **Mock High School National Exam**

Full Name in English: ..... | **Date of Examination: .....**

Room N°: ..... Seat N°: ..... | **Subject: Physics grade 12**

Date of Birth: ..... | **Duration: 50 mins**

✍ Accomplish great things take time. | **Scores: 75 marks**

✂.....

Noted: ▪ No materials, including books, documents, or cell phones, are permitted during exams.

▪ You're not allowed to cheat in an exam.

▪ A calculator is not permitted during an exam.

22

Questions and Exercises

① What is the function of a transformer? Describe its structure.

② How can the direction of the magnetic field be determined in:

(a) A circular current-carrying coil?

(b) A straight current-carrying wire?

③ Calculate the volume occupied by 9.6 grams of oxygen gas in a container at a pressure of 10^5 Pa and a temperature of 300 K. The universal gas constant is given as $R = 8.31$ J/mole. K, and the molar mass of oxygen gas is $M_{O2} = 32$ g/mole.

④ A cylindrical tube contains 4 moles of helium gas at a temperature of $27°C$. Assuming the gas behaves as an ideal gas with a molar mass of 4×10^{-3} kg/mol, determine:

(a) The total kinetic energy of the gas.

(b) The average kinetic energy of each molecule.

(c) The root-mean-square (rms) speed of each helium molecule.

(d) The temperature required for the rms speed to double, and the internal energy of the system at this new temperature.

⑤ In a sealed tube with a volume of $V = 20\text{mL}$, nitrogen gas with a mass of $m = 50$ mg is contained at a low temperature. Calculate the pressure of the nitrogen gas in the tube when the temperature is $T = 300\text{K}$. Assume the molar mass of nitrogen gas is $M = 28\text{g/mol}$ and the universal gas constant is $R = 8.31$ J/mol. K.

⑥ Imagine a Carnot engine that operates between the temperatures $T_h = 850$ K and $T_c = 340$ K. The engine performs 1500 J of work each cycle, which takes 0.25 s.

(a) What is the efficiency of this engine?

(b) What is the average power of this engine?

(c) How much energy Q_h is extracted as heat from the high-temperature reservoir every cycle?

(d) How much energy Q_c is expelled as heat to the low-temperature reservoir every cycle?

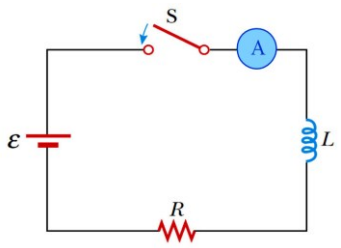
⑦ A circular 100-loop of wire, with its center at point O, has a radius of $R = 10$ cm and carries a current of $I = 10\text{A}$. Calculate the magnitude of the magnetic field (magnetic induction) at point O.

⑧ A solenoid has a length of 0.5 m and a diameter of 4 cm. It is wound with a wire of diameter 1mm in a tightly packed configuration. The wire has a resistivity of $\rho = 2.5\times10^{-7}$ Ωm.

(a) Sketch a diagram showing the direction of the current in the coil and the direction of the magnetic field.

(b) Calculate the resistance of the coil (neglecting the thickness of the insulation).

⑨ Consider the circuit shown in the figure. The electromotive force (emf) is $50.0\ \mathrm{V}$, the inductance is $25.00\mathrm{mH}$, and the resistance is $20.0\ \Omega$.



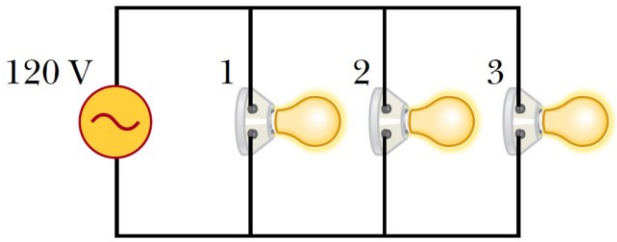
(a) Find the rate of change of current immediately after the switch is closed, and determine the current when the rate of change of current is $400\ \mathrm{A/s}$.

(b) Find the current through the circuit when the switch has been closed for a long time.

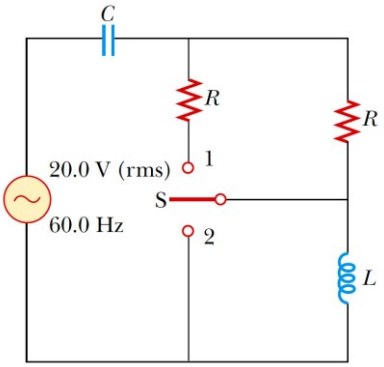
⑩ Figure below shows three lamps connected to a 120-V AC (rms) household supply voltage. Lamps 1 and 2 have 150-W bulbs; lamp 3 has a 100-W bulb.

(a) Find the rms current of each bulb.

(b) Find the resistance of each bulb.



⑪ A capacitor, a coil, and two resistors of equal resistance are arranged in an AC circuit, as shown in Figure P33.58. An AC source provides an emf of $20.0\ \mathrm{V}$ (rms) at a frequency of $60.0\ \mathrm{Hz}$. When the double-throw switch S is open, as shown in the figure, the rms current is $183\ \mathrm{mA}$. When the switch is closed in position 1, the rms current is $298\ \mathrm{mA}$. When the switch is closed in position 2, the rms current is $137\ \mathrm{mA}$. Determine the values of R, C, and L.



⑫ A section of copper wire is formed into a circular coil with a radius of $12\ \mathrm{cm}$. The coil's normal line is aligned parallel to a magnetic field that increases from $0\ \mathrm{T}$ to $0.60\ \mathrm{T}$ over a time interval of $0.45\ \mathrm{s}$. The copper wire has a resistance per unit length of $3.3 \times 10^{-2}\ \Omega/\mathrm{m}$. Calculate the electrical energy dissipated due to the resistance of the wire.

⑬ The wingspan (tip to tip) of a Boeing 747 jetliner is $59\ \mathrm{m}$. The plane is flying horizontally at a speed of $220\ \mathrm{m/s}$. The vertical component of the earth's magnetic field is $5.0 \times 10^{6}\ \mathrm{T}$. Find the emf induced between the plane's wing tips.